# सहायक प्राध्यापक (धातूशास्त्र)
## महाराष्ट्र अभियांत्रिकी शिक्षक सेवा, गट-अ
### उच्च व तंत्र शिक्षण विभाग
## Assistant Professor (Metallurgy)
## Maharashtra Engineering Teachers' Service, Group-A
### Higher and Technical Education Department

**परीक्षेचे टप्पे:** **१)** लेखी परीक्षा - २०० गुण **२)** मुलाखत- ५० गुण

## -: परीक्षा योजना :-

| विषय व सांकेतांक | माध्यम | प्रश्नसंख्या | गुण | कालावधी | दर्जा | प्रश्नपत्रिकेचे स्वरुप |
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| **विषयांशी संबंधीत घटक (११२५)** | इंग्रजी | १०० | २०० | एक तास | पदवी | वस्तुनिष्ठ बहुपर्यायी |

**अ) नकारात्मक गुणदान -**

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| १) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील. |
| २) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील. |
| ३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल. |
| ४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही. |

**ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.**

## -: अभ्यासक्रम :-

**विषयांशी संबंधित घटक या विषयांमध्ये खालील घटक व उपघटकांचा समावेश असेल.**

| अ.क्र. | घटक व उपघटक |
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| 1 | **Section 1: Engineering Mathematics** |
| | Linear Algebra: Matrices and Determinants, Systems of linear equations, Eigen values and Eigen vectors.<br>Calculus: Limit, Continuity and Differentiability; Partial derivatives; Maxima and minima; Sequences and series; Test for convergence; Fourier series.<br>Vector Calculus: Gradient; Divergence and Curl; Line, Surface and volume integrals; Stokes,Gauss and Green's theorems.<br>Differential Equations: Linear and non-linear first order ODEs; Higher order linear ODEs with constant coefficients; Cauchy's and Euler's equations; Laplace transforms; PDEs –Laplace, one dimensional heat and wave equations.<br>Probability and Statistics: Definitions of probability and sampling theorems, conditional probability, Mean, median, mode and standard deviation; Random variables; Poisson, normal and binomial distributions; Analysis of experimental data; linear least squares method.<br>Numerical Methods: Solutions of linear and non-linear (Bisection, Secant, Newton- Raphson methods) algebraic equations; integration by trapezoidal and Simpson's rule; single and multi-step methods for differential equations. |
| 2 | **Section 2: Metallurgical Thermodynamics and kinetics** |
| | Laws of Thermodynamics: First law – energy conservation, Second law - entropy; Enthalpy, Gibbs and Helmholtz free energy; Maxwell's relations; Chemical potential; Applications to metallurgical systems, solutions, ideal and regular solutions; Gibbs phase rule, phase equilibria, binary phase diagram and lever rule, free-energy vs. composition diagrams; Equilibrium constant, Activity, Ellingham and phase stability diagrams; Thermodynamics of point defects, surfaces and interfaces, adsorption and segregation phenomena.<br>Electrochemistry: Single electrode potential, Electrochemical cells, Nernst equation, Potential-pH diagrams.<br>Kinetics: First order reactions, reaction rate constant, Arrhenius relation, heterogeneous reactions, oxidation kinetics. |
| 3 | **Section 3: Transport Phenomena** |
| | Momentum Transfer: Concept of viscosity, shell balances, Bernoulli's equation, mechanical energy |

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| | balance equation, flow past plane surfaces and through pipes.<br>Heat transfer: Conduction, Fourier's Law, 1-D steady state conduction.<br>Convection: Heat transfer coefficient relations for forced convection.<br>Radiation: Black body radiation, Stefan-Boltzman Law, Kirchhoff's Law.<br>Mass Transfer: Diffusion and Fick's laws, Mass transfer coefficients.<br>Dimensional analysis: Buckingham Pi theorem, Significance of dimensionless numbers.<br>Electrochemical kinetics: Polarization |
| **4** | **Section 4: Mineral Processing and Extractive Metallurgy** |
| | Unit operations : Comminution techniques, Size classification, Flotation, Gravity and other methods of mineral beneficiation.<br>Agglomeration: Sintering, pelletizing and briquetting.<br>Metallurgical processes : Material and Energy balances<br>Principles and processes for the extraction of non-ferrous metals – Metallothermic and carbothermic reduction, application of Ellingham diagrams and solubility product for multicomponent system. Electrometallurgy and Pyrometallurgy processes, Rare Earth extraction metals.<br>Iron and steel making: Material and heat balance in blast furnace; Structure and properties of slags and molten salts – basicity of slags - sulphide and phosphate capacity of slags; Production of metallurgical coke.<br>Other methods of iron making (COREX, MIDRE)<br>Primary Steel Making: Basic oxygen furnace, process dynamics, oxidation reactions, electric arc furnace.<br>Secondary steel making: Ladle process – deoxidation, argon stirring, desulphurization, inclusion shape control, principles of degassing methods; Alloy steels and stainless steel manufacturing.<br>Continuous Casting: Fluid flow in the tundish and mould, heat transfer in the mould, segregation, inclusion control. |
| **5** | **Section 5: Physical Metallurgy** |
| | Chemical Bonding: Ionic, covalent, metallic, and secondary bonding in materials, Crystal structure of solids – metals and alloys, ionic and covalent solids, and polymers.<br>Characterization Techniques : X-ray Diffraction – Bragg's law, optical metallography, principles of SEM and TEM imaging. Mechanical testing<br>Crystal Imperfections: Point, line and surface defects; Coherent, semi-coherent and incoherent interfaces.<br>Diffusion in solids: Diffusion equation, steady state and error function solutions; Examples-homogenenization and carburization; Kirkendall effect; Uphill diffusion; Atomic models for interstitial and substitutional diffusion; Pipe diffusion and grain boundary diffusion.<br>Phase Transformation: Driving force, Homogeneous and heterogeneous nucleation, Growth Kinetics Solidification in isomorphous, eutectic and peritectic systems, cast structures and macrosegregation, dendritic solidification and constitutional supercooling, coring and microsegregation.<br>Solid State Transformations: Precipitation, spinoidal decomposition, ordering, massive transformation, discontinuous precipitation, eutectoid transformation, diffusionless transformations; Precipitate coarsening, Gibbs-Thomson effect.<br>Heat Treatment : Principles of heat treatment of steels, TTT and CCT diagrams; Surface hardening treatments; Recovery, recrystallization and grain growth; Heat treatment of Alloys steels, HSS, cast iron and nonferrous alloys.<br>Materials Science : Electronic, magnetic and optical properties of materials.<br>Corrosion : Basic forms of corrosion and its prevention |
| **6** | **Section 6: Mechanical Metallurgy and Non Destructive testing** |
| | Strength of Materials : Strain tensor and stress tensor, Representation by Mohr's circle, elasticity, stiffness and compliance tensor, Yield point, Plastic deformation by slip and twinning.<br>Dislocation theory: Edge, screw and mixed dislocations, source and multiplication of dislocations, stress fields around dislocations; Partial dislocations, dislocation interactions and reactions.<br>Strengthening mechanisms: Work/strain hardening, strengthening due to grain boundaries, solid solution, precipitation and dispersion.<br>Fracture : Fracture behaviour, Griffith theory, linear elastic fracture mechanics, fracture toughness, fractography, ductile to brittle transition.<br>Fatigue: Cyclic stress strain behaviour - low and high cycle fatigue, crack growth. Mechanisms of high temperature deformation and failure; creep and stress rupture, stress exponent and activation energy.<br>Non-destructive Testing (NDT) : Dye-penetrant, ultrasonic, radiography, eddy current, acoustic emission and magnetic particle inspection methods. |

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| 7 | **Section 7: Manufacturing Processes** |
| | Metal casting: Mould design involving feeding, gating and risering, casting practices, casting defects.<br>Secondary operations: Metal forming – fundamentals of metal forming processes of rolling, forging, extrusion, wire drawing and sheet metal forming, defects in forming.<br>Metal joining: Principles of soldering, brazing and welding, welding metallurgy, defects in welded joints in steels and aluminium alloys.<br>Powder Metallurgy: Production of powders and properties, compaction and sintering and advanced sintering processes. |

दिनांक – ३० एप्रिल, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग